

# GLOSSARY

## A

### Absolute magnitude

A measure of the true brightness of a star or other astronomical object, defined as the apparent magnitude it would have at a distance of 10 parsecs (32.6 light-years). See also *apparent magnitude*, *luminosity*

### Accretion

(1) The colliding and sticking together of small astronomical bodies to make larger ones. (2) The process whereby a body grows in mass by accumulating matter from its surroundings.

### Active galaxy

A galaxy that emits an exceptional amount of electromagnetic radiation over a wide range of wavelengths. The radiation comes from a central “active galactic nucleus” and is thought to be powered by the accretion of gas onto a supermassive black hole. There are several named types, although the apparent differences between them may be because we are viewing them at different angles as seen from Earth. See also *blazar*, *quasar*, *Seyfert galaxy*

### Apparent magnitude

A measure of the brightness of a star or other astronomical object as seen from Earth, which depends on its closeness as well as how luminous it really is. The brighter the object, the smaller the numerical value of its apparent magnitude. See also *absolute magnitude*, *luminosity*

### Asterism

A pattern of bright stars in the night sky that is usually only part of a constellation. For example, the Big Dipper (or Plough) is an asterism within the constellation Ursa Major. See also *constellation*

### Asteroid

A small, irregularly shaped Solar System object of rock or metal less than 600 miles (1,000 km) in diameter. Most, but not all, asteroids are found in the Asteroid Belt between Mars and Jupiter.

### Astronomical unit (AU)

A unit of distance based on the average distance between Earth and the Sun. It is approximately 93 million miles (150 million km).

### Atmosphere

A gaseous surround to a planet or a low-density region of plasma around a star.

### Atom

A building block of ordinary matter. It consists of a central heavy nucleus of protons (positively charged, with a different number for each chemical element) and neutral neutrons, surrounded by orbiting electrons (negatively charged, and equal in number to the protons). See also *electron*, *ion*

### Aurora

A display of glowing light in the sky that is most common in polar regions. It is caused by high-energy particles from the Sun being deflected by Earth's magnetic field and colliding with atoms in Earth's atmosphere.

### Axis

The imaginary line about which a body rotates.

## B

### Background radiation

See *cosmic microwave background radiation*

### Barred spiral galaxy

A galaxy with spiral arms that originate from the ends of an elongated, bar-shaped central region. See also *spiral galaxy*

### Big Bang

The event in which the Universe was born. According to Big Bang theory, the Universe originated around 13.8 billion years ago in an extremely hot dense state and has been expanding ever since.

### Binary star

A pair of stars that orbit each other around a common center of mass. See also *center of mass*

### Black dwarf star

A former white dwarf star, which has cooled so much that it emits no detectable light. The Universe is not yet old enough for black dwarfs to have formed. See also *brown dwarf star*, *white dwarf star*

### Black hole

A region of space into which so much matter has collapsed that nothing, not even light, can escape its gravitational pull. The supermassive black holes found in the centers of galaxies can be up to billions of times the mass of the Sun.

### Blazar

The most luminous and variable type of active galaxy in terms of its radiation. See also *active galaxy*

### Blueshift

The opposite of redshift: the shifting of electromagnetic radiation to a higher frequency when it radiates from an object moving toward the observer. See also *redshift*

### Bok globule

A type of compact dark nebula believed to be the precursor of a protostar. See also *protostar*

### Brown dwarf

A body that forms out of a contracting cloud of gas, like a star, but because it contains too little mass never becomes hot enough to sustain nuclear fusion.

## C

### Celestial equator

An imaginary circle on the celestial sphere that is a projection of the Earth's own equator onto it. See also *celestial sphere*

### Celestial poles

The points in the sky directly above Earth's north and south poles. The celestial sphere appears to rotate around an axis joining the celestial poles.

### Celestial sphere

An imaginary sphere that surrounds the Earth. As the Earth rotates from west to east, the sphere appears to rotate from east to west. In order to define the position of stars and other celestial bodies, it is convenient to think of them as being attached to the inside surface of this sphere.

### Center of mass

The point around which two or more bodies revolve, as for example when two stars revolve around each other. If one of the bodies has more mass than the other, the center of mass lies toward the larger one.

### Cepheid variable

A type of variable star whose luminosity alters in a regular rhythm. Cepheids vary in brightness as they physically expand and contract. The more luminous the Cepheid, the longer its period of variation. See also *luminosity*, *variable star*

### Chromosphere

The relatively thin layer of the Sun's atmosphere that lies between the photosphere and the corona. See also *corona*, *photosphere*

### Comet

A small body composed mainly of dust-laden ice that orbits the Sun. When a comet enters the inner Solar System some of its material evaporates, often forming a long “tail” of gas and dust.

### Constellation

(1) A named pattern of stars in the night sky, used for convenience as a way of describing the position of astronomical objects seen from Earth. (2) One of the 88 regions into which the celestial sphere is divided for reference purposes, based on these traditional constellations.

### Corona

The outermost part of the atmosphere of the Sun or another star, stretching thousands of miles into space. It has a very high temperature but a low density.

### Cosmic microwave background radiation (CMBR)

The radiation left over from the Big Bang, appearing from all directions of the sky.

## D

### Dark energy

A little-understood phenomenon that appears to account for about 70 percent of the total “mass plus energy” in the Universe. It is thought necessary to explain why the expansion of the Universe is currently accelerating.

### Dark matter

A mysterious kind of matter that seems to interact only via gravity and not by emitting or absorbing electromagnetic radiation, in contrast to ordinary matter made of atoms. Scientists think it exists in large quantities in the Universe because without it, galaxies should fly apart as they rotate.

### Declination

The equivalent on the celestial sphere of latitude on Earth. The declination of a star is its angular distance north or south of the celestial equator. See also *right ascension*

### Diffuse nebula

A nebula lacking sharp outer boundaries and without obvious internal features. See also *nebula*

### Double star

Two stars that appear close together in the sky. If they actually orbit each other, the system is called a binary. An optical double consists of stars that